

APPENDIX A. TRAINING AND IMPLEMENTATION DETAILS

For WTB wake supervision,

*Training loop summary***Algorithm A1** Physics-aligned MoE training

Input: history windows X , graph sequence A , targets Y , primary regime anchors $\mathbf{p}_i$, auxiliary wake labels W (WTB only), model parameters θ

for each minibatch do

Build node-level graph context from the precomputed graph tensors

Encode X with the directed-diffusion GRU to obtain $\mathbf{z}_{i,t}$ where Ω_{wake} contains only MPPT and pitch-control samples

Compute gate logits Z and soft routing probabilities $\mathcal{A}_{i,j}$ with defined wake flags. The graph-smoothness penalty is

Compute expert forecasts and aggregate them with G

Evaluate L_{pred}

If MoE mode is active:

add L_{bal} on top-K routing statistics

add L_{align} on valid primary regime anchors

if WTB: add L_{force} on MPPT/pitch samples

if WTB: add L_{aux} on wake labels

add L_{smooth} on the final graph snapshot

Update θ with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) + \epsilon}.$$

Graph construction details

For WTB, the downstream unit vector is $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$. With $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$, the streamwise and cross-stream distances are $d_{ij,t}^\parallel = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$ and $d_{ij,t}^\perp = |\Delta \mathbf{p}_{ij}^\perp \mathbf{u}_{i,t}|$. A candidate edge activates when

Model-family and supplementary evidence index

Table A1 is designed to separate shared representation capacity from expert allocation, following the standard mixture-of-experts comparison logic [Jacobs1991adaptive; @shazeer2017outrageously].

TABLE I

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Table A1. In-family model comparison used for mechanism validation. Encoder capacity is held approximately fixed so that routing terms, rather than parameter count alone, explain the mechanism contrast.

Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110.012 / ERA5 103.272	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110.012 / ERA5 103.843	Router-capacity control
Corrected routing comparator	Node-level soft gate	WTB boundary-forced: $L_{\text{hard}} + L_{\text{align}} + L_{\text{force}}$; ERA5: full corrected stack	WTB 110.012 / ERA5 103.843	Accountability comparator

Information boundary and channel roles

Table A2 makes the leakage and shared-anchor boundary explicit. The routing labels and some gate anchors deliberately share wind-speed and pitch information; the reported NMI/ARI therefore audits compliance with a declared SCADA boundary, not anchor-free discovery.

TABLE II

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Table A2. Information boundary and channel-role audit.

Quantity	Label construction	Model input / gate anchor	Supervised target	Timing boundary
Wepd	WTB regime label	history and issue-time anchor	no	$\leq t$ only
Pab_mean	WTB regime label	history and issue-time anchor	no	$\leq t$ only
PatV _{t-4t+1:t}	no	historical input / issue-time status	no	$\leq t$ only
PatV _{t+1:t+P}	no	no	yes	future target only
Wake score	auxiliary wake flag	graph-derived anchor	no	computed from issue-time graph
Declared regime label	alignment/forcing supervision	no test-time input	no	train/validation supervision only
Confirmed threshold stream	degraded-rule comparator	not a model input	no	delayed/missing/noisy in audit

Auxiliary losses

Let $\mathcal{B}$ denote the routed samples after masks, with $B = |\mathcal{B}|$. The soft importance and normalized share of expert e are $I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}$ and $P_e = I_e / \sum_r I_r$. The top-K load is $f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)]$, giving

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

where σ is the median retained neighbor distance on the training graph.

$$W_{i,t} = \begin{cases} 1, & s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathcal{A}(i,j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

Shared constants

TABLE III

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Table A3. Shared architecture, graph, and training constants used in the reported experiments. Shared constants are listed before dataset-specific values so that controlled factors can be distinguished from observability-specific design choices.

Item	WTB	ERA5	Role
History / horizon (H, P)	(36, 24)	(36, 24)	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top- K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius d_{max}	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

Dataset-specific thresholds and routing weights

TABLE IV

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Table A4. Dataset-specific regime thresholds used to construct routing anchors. The values are operational definitions fixed or estimated before test evaluation; they are not universal turbine constants.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s ⁻¹ , idle / active split
WTB	u_{rated}	10.5 m s ⁻¹ , MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of $ \text{sshf} $ on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split

The loss-weight table follows the physics-informed learning principle that a scientific constraint must be named and scaled explicitly [[@karniadakis2021piml](#); [@karpatne2017tgd](#)s].

TABLE V

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Table A5. Active routing-loss weights in the reported corrected models. The table records which regulariser is active in each observability setting.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence

Statistical claim boundaries

Table A6 separates the train-only RMSE guardrail from paired seed-level tests on the train-only checkpoints. The iTransformer row is the displayed five-seed train-only guardrail gap; the Graph WaveNet row is a legacy pure-prediction baseline contrast; the gate-versus-full-MoE rows use the official reviewer-stat-pack paired statistics on train-only checkpoints for both families.

TABLE VI
***Table A6.** Statistical claim boundaries used for reviewer-facing wording.

Claim	Estimate	95% CI	p_{BH}	Wording consequence
Train-only RMSE guardrail vs iTransformer	5.589	–	–	Displayed guardrail gap for the provenance-corrected rerun
Legacy RMSE price vs Graph WaveNet	9.029	–	<0.001	Legacy historical checkpoint contrast only
Legacy boundary-band RMSE price	17.758	–	0.001	Boundary-window price for the legacy historical checkpoint only
Gate NMI vs full physics-aligned MoE (train-only)	-0.109	[-0.280, 0.084]	0.352	No significant alignment difference; do not claim boundary forcing is required for alignment
Gate ARI vs full physics-aligned MoE (train-only)	-0.138	[-0.356, 0.088]	0.379	Same; no superiority wording
Boundary quantile cost vs GWN physical bin	-0.263M	[-10.704M, 9.782M]	–	CI crosses zero; reserve cost should remain a diagnostic claim
Boundary quantile violation vs GWN physical bin	0.003	[-0.011, 0.020]	–	CI crosses zero; no universal reserve-policy optimality claim

Class-weight sensitivity audit

Table A6b makes the class-weight provenance boundary explicit. The legacy strict-cache row records the historical boundary-router checkpoint from earlier iterations; the train-only rerun recomputes alignment and pitch-forcing class weights from the training split only and serves as the single source for all headline audits in the main paper. The train-only rerun lowers NMI/ARI but remains above the routing-claim threshold.

TABLE VII
***Table A6b.** Class-weight sensitivity audit. Boundary-router rerun after recomputing alignment and pitch-forcing loss weights from the training split only.

Condition	Overall RMSE	Switch RMSE	Pitch RMSE	NMI	ARI
Legacy strict-cache weights	236.13	239.86	286.95	0.8716	0.9166
Train-only weight rerun	229.93	231.65	289.63	0.7208	0.7398
Delta	-6.20	-8.21	2.68	-0.1508	-0.1768

Outcome-channel sanity audit

Table A7 adds a bounded check for the shared-anchor concern. It does not use pitch-threshold labels to score the contrast. Validation data define a wind-speed-bin power curve, and the test comparison is restricted to 9.5–11.5 m s⁻¹ boundary anchors with fine wind-bin adjustment. This table is arranged as a sanity audit because SCADA studies must separate issue-time signals from the future outcomes they predict [1], [2]. The result asks whether the recovered gate separates samples with different future active-power response, not whether it discovers a regime without anchors.

TABLE VIII
***Table A7.** Outcome-channel sanity audit for the shared-anchor concern.

Check	Five-seed summary	Interpretation
Boundary cells per seed	3276 MPPT / 10465 pitch	9.5–11.5 m s ⁻¹ test anchors
Future mean power	100.0 ± 17.5 kW	Pitch-gate minus MPPT-gate after wind-bin adjustment
Power-curve residual	100.0 ± 17.5 kW	Validation wind-bin power curve only; no pitch label used
Future power ramp	81.8 ± 24.7 kW	Mean late-horizon minus early-horizon power
Anchor-time Patv	-44.1 ± 32.5 kW	Current active power is not driving the same positive contrast
Residual claim boundary	sanity check only	Mitigates circularity concern; does not prove anchor-free discovery

External-site deployment-gate audit

Table A8 records the deployment interpretation used for the external evidence. The first row is the cross-site mechanism replication: retrained on the ENGIE La Haute Borne farm, where blade pitch is directly observed in 99.2% of cells, the routing mechanism recovers the declared boundary and clears the held-out criterion. The remaining rows map the Kelmarsh/Penmanshiel signals to the deployment action that follows. The table is deliberately decision-shaped, with evidence, criterion and consequence in separate columns. The signature-gate probe (Table A9d) upgrades the binary go/no-go into a graded signature-strength reading: partial pitch observability modulates rather than extinguishes boundary recoverability.